# Blessed G/Michael Catholic Secondary School

## Semester 1 IT Mid1 Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

**Name:** _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

**I. Choose the best answer among the given alternatives.**

____ **1.** In asymmetric public-key cryptography, a message encrypted with a recipient's public key can only be decrypted by:

**A.** The sender's public key **B.** The recipient's private key

**C.** The sender's private key **D.** A symmetric secret key

____ **2.** Which software development methodology emphasizes short iterative sprints, frequent releases, and continuous stakeholder feedback?

**A.** Waterfall model **B.** Agile methodology

**C.** V-Model **D.** Linear Sequential Model

____ **3.** Which of the following hash algorithms produces a 256-bit fixed-length cryptographic digest?

**A.** MD5 **B.** SHA-1 **C.** SHA-256 **D.** DES

____ **4.** Which cryptographic mechanism provides non-repudiation, message integrity, and sender authentication?

**A.** Digital signature **B.** Symmetric DES encryption

**C.** Caesar cipher **D.** Base64 encoding

____ **5.** In software engineering, what is the primary drawback of the traditional Waterfall development model?

**A.** Inflexibility in accommodating changes once requirements are locked
**B.** Excessive communication with end users
**C.** Lack of documentation
**D.** Unplanned daily sprint meetings

____ **6.** Which database ACID property guarantees that all transactions are committed permanently even in the event of power outage?

**A.** Durability **B.** Atomicity **C.** Consistency **D.** Isolation

____ **7.** What is the time complexity of searching for an element in a balanced Binary Search Tree (BST) with N nodes?

**A.** O(log N) **B.** O(N) **C.** O(1) **D.** O(N^2)

____ **8.** Which type of software testing verifies that individually developed units and modules operate together harmoniously?

**A.** Integration testing **B.** Unit testing

**C.** Regression testing **D.** Stress testing

____ **9.** Which protocol secures web browser sessions by encrypting data packets via TLS at the transport layer?

**A.** HTTPS **B.** FTP **C.** Telnet **D.** SNMP

____ **10.** What is a Distributed Denial of Service (DDoS) attack?

**A.** An attack overwhelming a target server with flood traffic originating from multiple compromised devices (botnet)
**B.** An attack performed by a single modem
**C.** A database query optimization technique
**D.** An optical cable malfunction

____ **11.** In object-oriented programming, the principle where a subclass can provide a specialized implementation of a method defined in its superclass is:

**A.** Polymorphism (Method Overriding) **B.** Encapsulation

**C.** Abstraction **D.** Compilation

____ **12.** Which symmetric key encryption standard is widely used globally by governments with key lengths of 128, 192, or 256 bits?

**A.** AES (Advanced Encryption Standard) **B.** DES

**C.** ROT13 **D.** MD5

____ **13.** What is a zero-day vulnerability in cybersecurity?

**A.** A software security flaw that is unknown to the vendor and has no available patch

**B.** A flaw discovered on the first day of each month

**C.** A virus that takes zero days to write

**D.** A password with zero characters

____ **14.** In network routing, which metric does the RIP (Routing Information Protocol) algorithm use to select paths?

**A.** Hop count **B.** Bandwidth **C.** Link delay **D.** Reliability

____ **15.** Which normal form eliminates transitive functional dependencies where a non-key attribute depends on another non-key attribute?

**A.** Third Normal Form (3NF) **B.** First Normal Form (1NF)

**C.** Second Normal Form (2NF) **D.** Unnormalized Form

____ **16.** Which device operates at Layer 3 of the OSI model to inspect IP addresses and interconnect different subnets?

**A.** Router **B.** Repeater **C.** Hub **D.** Bridge

____ **17.** What is SQL Injection (SQLi)?

**A.** A web vulnerability where malicious SQL commands are inserted into input fields to manipulate backend databases

**B.** A technique to speed up database queries

**C.** An automated backup tool

**D.** A hardware failure in database servers

____ **18.** Which data structure follows the First-In, First-Out (FIFO) processing order?

**A.** Queue **B.** Stack **C.** Tree **D.** Graph

____ **19.** What is the role of an Operating System Kernel?

**A.** It is the core program that directly manages CPU scheduling, memory allocation, and hardware device drivers

**B.** It is the desktop wallpaper software

**C.** It is the internet browser engine

**D.** It is the peripheral power cable

____ **20.** Which cloud deployment model is provisioned for exclusive use by a single organization comprising multiple consumer business units?

**A.** Private cloud **B.** Public cloud

**C.** Community cloud **D.** Hybrid cloud